

Erratum: The entropy-gap formula of Section 8.3 is incorrect, and the entropic estimate of the conformal dimension is withdrawn

Erratum to “The Canonical Form of Attention: Positive Geometry, SYK Vertices, and Superconformal Symmetry in Transformer Architectures” (DOI 10.5281/zenodo.18971720, March 11, 2026)

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Summary

Section 8.3 of the paper asserts that a power-law attention distribution $\alpha(r) \propto r^{-2\Delta}$ on n elements has entropy gap

$$H_{\text{gap}} = 2\Delta \cdot \log n + \text{const.} \quad (8.3)$$

This formula is incorrect. For a normalized power law with exponent $s = 2\Delta < 1$, the entropy gap converges to a **constant**; its slope against $\log n$ over the paper’s own measured range is an order of magnitude smaller than formula (8.3) predicts. Consequently the paper’s entropic estimate of the conformal dimension, $\Delta_{\text{eff}} = 0.254$, and the headline claim that *two independent observables measure one exponent to 1.4%*, are **withdrawn**. The entropy-gap *measurements* themselves stand; what falls is the inference from them to Δ . The paper’s exact results (the canonical-form identity, the quartic vertex, the perturbative identity $H_{\text{gap}} = \frac{1}{2}\text{Var}(s)$) and its direct power-law profile measurements are unaffected.

1. The correct algebra

Let $\alpha(r) = r^{-s}/Z_n$ on $r \in \{1, \dots, n\}$ with $0 < s < 1$. By Euler–Maclaurin,

$$Z_n = \sum_{r=1}^n r^{-s} = \frac{n^{1-s}}{1-s} + \zeta(s) + O(n^{-s}),$$

and the entropy decomposes as $H = \log Z_n + s \mathbb{E}[\log r]$, with

$$\mathbb{E}[\log r] = \log n - \frac{1}{1-s} + o(1).$$

Therefore

$$H = \log n - \left[\log(1-s) + \frac{s}{1-s} \right] + o(1), \quad H_{\text{gap}}(n) \equiv \log n - H \longrightarrow \log(1-s) + \frac{s}{1-s},$$

a constant. For $s = 1/2$ the limit is $1 - \log 2 \approx 0.307$ nats. The gap grows logarithmically only for $s \geq 1$ (slope $\rightarrow 1/2$ at the Zipf point $s = 1$, slope $\rightarrow 1$ for $s > 1$) — never with slope s .

Origin of the error. Formula (8.3) is what results if the energy term $s \mathbb{E}[\log r] \approx s(\log n - \text{const})$ is dropped from $H = \log Z_n + s \mathbb{E}[\log r]$: the truncated expression $H \approx \log Z_n \approx (1 - s) \log n$ gives a spurious gap slope of s . The dropped term restores the full $\log n$ in H .

2. Numerical demonstration, on the paper’s own range

Exact entropies of the normalized power law, ordinary-least-squares slope of H_{gap} against $\log n$ over $n \in [4, 256]$ (the range used in Section 8):

$s = 2\Delta$	Measured gap slope, $n \in [4, 256]$	Formula (8.3) predicts	Asymptotic gap (constant)
0.3	0.011	0.30	0.072
0.5	0.041	0.50	0.307
0.7	0.102	0.70	1.129
1.0	0.264	1.00	$\sim \frac{1}{2} \log n$

Even at finite range the discrepancy is an order of magnitude, and **no normalized power law of any exponent reproduces the measured gap slope of 0.507** — the steepest case, $s = 1$, reaches only 0.264 on this range. The computation is elementary and self-contained; the generating script (`k1_powerlaw_gap.py`, ~50 lines, pure math, no model or data dependencies) is included with this erratum.

3. Statements affected

The following claims of the published paper are withdrawn or corrected:

- **Contributions, items 6–7 and 15; Abstract statements derived from them.** The identification of the gap slope with 2Δ , the value $\Delta_{\text{eff}} = 0.254$, the 1.4% agreement with SYK₄’s $\Delta = 1/4$, and the claim that the entropy gap provides “a stable integral measurement” of the conformal dimension.
- **Section 8.3 in full.** The formula, the $a = 2\Delta$ consistency table (measured $a = 0.507$, $\beta = 0.511$, $\Delta_{\text{eff}} = 0.254$), and the claim that the entropy gap is a cleaner estimator of Δ than direct profile fitting.
- **Section 8.4.** The derivation $k_{\text{eff}} \propto n^{1-2\Delta}$ rests on the same algebra and is incorrect for $s < 1$ (a normalized power law with $s < 1$ has $k_{\text{eff}} \propto n$ up to constants). The measured $\beta = 0.533$ stands as a measurement but is not a measurement of $1 - 2\Delta$.
- **Section 7.4’s resolution paragraph and the corresponding statements in the Discussion and Conclusion** (“the dimension IS accessible ... through the entropy gap”; “confirmed by two independent measurements”; “ $H_{\text{gap}} = 0.507 \cdot \log n$, matching the Calabrese–Cardy entanglement entropy formula” as an inference about Δ). The negative result of Section 7.4 — that the conformal dimension is not accessible through the observables of this paper — stands unresolved rather than partly resolved.
- **The effective central charge $c_{\text{eff}} = 3a_L$ of Section 8.2**, as an inference: the identification of the measured logarithmic slopes with a Calabrese–Cardy central charge ran through the same class of reasoning and should be treated as an uninterpreted empirical slope pending an actual derivation.

4. What is unaffected

- The exact canonical-form identity (Sections 2–4) and its numerical verification, including the quartic (σ^4) vertex result.
- The perturbative identity $H_{\text{gap}} = \frac{1}{2}\text{Var}(s)$ (Section 8.1), verified to four significant figures — a different regime, not derived from formula (8.3).
- **The measurements of Section 8 as measurements.** The logarithmic growth of the trained-model entropy gap ($R^2 > 0.97$ across layers, $R^2 = 0.992$ at Layer 0) and the measured k_{eff} scaling are real and reproducible. What this erratum removes is their interpretation as a measurement of the conformal dimension.
- The direct power-law profile measurements (Sections 7 and 8.6), with the length-dependence caveats the paper already states.

5. What the entropy gap does measure

With the correct algebra, a logarithmically growing gap cannot come from a normalized power-law profile of any exponent; it indicates **n -independent concentration structure** — a component of attention mass whose size does not scale with context length (for a distribution placing an n -independent mass fraction p on a bounded set, the gap slope is $\approx p$, suggesting $p \approx 0.5$ at this paper’s protocol). The decomposition of the measured profiles into scaling and localized components is ongoing follow-up work and is deliberately not asserted here beyond the sign of the conclusion: the gap slope measures concentration, not the window exponent.

6. Provenance

The error was found on August 8–9, 2026, during a pre-registered re-derivation session: the claim that formula (8.3) is incorrect, the suspected mechanism (the dropped energy term), and the kill condition (slope ≥ 0.25 at $s = 0.5$ would falsify the correction) were committed in writing before the numerics were run. The check confirmed the correction (measured slope 0.041). This erratum is issued under the program’s standing method: corrections are published at the same prominence as results.